Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “superposition” and “interference” to compare your model or result. Sentence starters are optional.

superposition: adding displacements or fields at the same place and time

interference: the pattern produced by overlapping waves

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

Across a stationary boundary, frequency remains 3 Hz while speed changes from 15 to 21 m/s.
Find both wavelengths.

Word help: boundary = the place where a medium or constraint changes

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add signed displacements; label units.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two upward pulses overlap.
Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

Two coherent sound waves have equal amplitude A at a microphone.
Draw their contributions and resultant when phase difference is 0°, then 180°.
Compare resultant amplitudes. What must stay stable for the pattern to persist?

Sentence starter: Both ... . They differ in ... because ... .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A process has ΔU=-24 J and W_{by}=32 J.
Find Q and decide whether heat enters or leaves.
Label energy entering, leaving and stored.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Three instantaneous displacements are +5,-2,-6 mm. **Find** resultant and state whether this single value determines all wave energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two waves at a point have equal amplitudes 4 cm. **Find** maximum constructive amplitude and complete destructive amplitude; explain the required relative phases.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3